Diagnosing Separation and Initial-Condition Errors

Separating the variables before integrating, carrying the constant of integration correctly, and applying the initial condition at the right step

Directions. Separable equations go wrong in three predictable places. Watch for all three below.

- integrating before the variables are actually separated;
- losing the constant of integration, or adding it at the wrong moment;
- applying the initial condition too early or too late.

Where a student's work is shown, **(i)** name the first wrong line, **(ii)** say what rule it breaks, **(iii)** solve correctly. Round decimal answers to two places, and check each solution by substituting it back into the differential equation.

1. Find the general solution of $\dfrac{dy}{dx} = 8x^3 - 6x$. Include the constant of integration, and say in one phrase what that constant represents on a slope field.

Answer: ________________

2. The same equation, $\dfrac{dy}{dx} = 8x^3 - 6x$, now carries the initial condition $y(1) = 4$. Find the particular solution, then evaluate $y(2)$. State the step at which the initial condition may first be used.

Answer: ____________

3. **Find and fix.** Solve $\dfrac{dy}{dx} = 2xy$ with $y(0) = 3$, then find $y(1)$. Wren's work:

$$\boxed{\int dy = \int 2xy\,dx \;\Rightarrow\; y = x^2 + C; \quad y(0) = 3 \Rightarrow C = 3; \quad y(1) = 4}$$

Name the first wrong line, say what Wren did to the y on the right side, and give the correct $y(1)$ to two decimal places.

Answer: ____________

4. Find and fix. Solve $\dfrac{dy}{dx} = 5y$ with $y(0) = 6$, then find $y(1)$. Dev's work:

$$\boxed{\ln|y| = 5x + C \;\Rightarrow\; y = e^{5x} + C; \quad y(0) = 6 \Rightarrow C = 5; \quad y(1) \approx 153.41}$$

Dev's first line is correct. Name the line that is not, explain what exponentiating does to a *sum*, and find $y(1)$ correctly.

Answer: ____________________

5. Find and fix. Solve $\dfrac{dy}{dx} = \dfrac{x}{y}$ with $y(0) = 4$, then find $y(3)$. Rosa separates correctly and reaches $\dfrac{y^2}{2} = \dfrac{x^2}{2} + C$, then writes:

"take the square root of both sides": $y = x + C$; $\quad y(0) = 4 \Rightarrow C = 4$; $\quad y(3) = 7$

Say what is wrong with that square root, then solve correctly for $y(3)$.

Answer: __________________

6. **Find and fix.** Solve $\dfrac{dy}{dx} = y^2$ with $y(0) = 2$, then find $y(0.25)$. Sam's work:

$$\boxed{\int y^{-2}dy = \int dx \;\Rightarrow\; \frac{1}{y} = x + C; \quad y(0) = 2 \Rightarrow C = 0.5; \quad y(0.25) \approx 1.33}$$

One sign is missing. Fix it and give $y(0.25)$. Then say what the corrected solution does as x approaches 0.5, and why that behaviour is invisible in Sam's version.

Answer: __________________

7. Solve $\dfrac{dy}{dx} = \dfrac{6x}{x^2+1}$ with $y(0) = 2$ and evaluate $y(2)$ to two decimal places. Then explain why applying the initial condition *before* integrating would be meaningless, and what answer a student who forgets the constant of integration entirely would report.

Answer: __________________

8. A classmate argues: "A constant is a constant — it makes no difference whether you add it before or after exponentiating." Using $\frac{dy}{dx} = 5y$ from problem 4, refute this. Differentiate $y = e^{5x} + 5$, compare the result with $5y$, and explain what the constant becomes when it is carried through the exponential correctly.